COMP 9200
Applied Research Methods

Finding a research problem

Introduction

Three comments reviewers write over and over

- **“I can’t tell what is new here.”**
  - An area and a plan to build something, but no claim that could turn out to be false.
- **“Why would anyone care about this result?”**
  - The motivation is that the topic is popular, not that a decision or design would change if the answer were known.
- **“This is a build plan, not a research plan.”**
  - Activities are listed (implement, test, deploy) but nothing could distinguish success from failure.

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Topic vs problem vs question

	What it is	Example
Topic	An area. Nothing to be right or wrong about.	Federated learning on edge devices
Problem	A specific failure, limitation or unknown in current practice, with a reason it exists.	Accuracy drops sharply when client data are non-IID. Existing corrections assume the server knows each client's label distribution, which it cannot observe.
Question	Something answerable with evidence, naming the comparison and the setting.	Can the server estimate label skew from update statistics alone, and does correcting with that estimate recover accuracy relative to FedProx at the same communication budget?

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Engineering task or research problem?

Engineering task

Build a mobile app that shows campus parking occupancy from crowd-sourced reports.

Research problem

Crowd-sourced occupancy prediction is accurate in studies that pay reporters. Whether accuracy survives when incentives are removed is unknown.

Four tests

1. Is the outcome uncertain? If you already know it will work, it is a task.
2. Would a negative result still teach the field something?
3. Would someone outside your group want to cite the answer?
4. Is success established by evidence, or just by effort and delivery?

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Where research problems come from

1. Limitations sections

Recent papers tell you what they could not do. Read the last two pages first.

2. Assumptions that break

A method assumes IID data, ample memory, a benign user, a fixed schema. Change the setting; does it still hold?

3. Contradictory findings

Two papers disagree. Which conditions explain the difference?

4. Technique transfer

A method from PL, databases or HCI has never been tried in your subfield. Why not, and would it survive?

5. Measurement gaps

A belief everyone repeats but nobody has measured. Characterization studies are cheap and citable.

6. Practitioner pain

Something you or an internship team worked around by hand. Ask why no tool exists.

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FINER method

From clinical research (Hulley et al.), read for computer science

	Criterion	What it means for a CS Master's project
F	**Feasible**	Data you can get, compute you have access to, skills you hold or can acquire in weeks, time left in the program.
I	**Interesting**	To you, because you will live with it for a while; and to a community, because a venue publishes this kind of result.
N	**Novel**	Not already answered. Check arXiv, DBLP, Google Scholar – do your lit review.
E	**Ethical**	Consent for data, dual-use potential, human-subjects approval where needed, environmental cost of compute.
R	**Relevant**	Someone would change what they do based on the answer. Name that person or role.

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Heilmeier questions

Question	How proposals fail it	Fix
Q1 What are you trying to do? No jargon.	*"We propose a contrastive self-supervised framework for cross-modal alignment."*	Rewrite so a smart friend in another department can repeat it back.
Q2 How is it done today, and what are the limits?	*The proposal skips straight to the new idea; the reviewer suspects the writer does not know current practice.*	Two honest paragraphs on the state of the art and exactly where it stops working.
Q4 Who cares? What difference will it make?	*"This is an active and important area."*	Name a decision, a design or a belief that changes if the answer is yes. Name who holds it.
Q8 What checks tell you whether it is working?	*The only check is the final result.*	A checkpoint at month two that could already tell you the approach is failing.

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Find the gap with a comparison table

Two recent surveys → **Snowball references** → **Skim 15, read 5** → **One row per paper**

Paper	Approach	Key assumption	Evaluation setting	Reported limitation
Li 2024	Server-side reweighting	Server knows client label counts	CIFAR-10, 100 clients, simulated	Fails when counts are private
Okafor 2025	Client-side regulariser	Clients have spare compute	FEMNIST, 50 clients	Not tested under stragglers
Your gap	—	Server sees only updates	?	?

A real gap is a pattern of empty cells

Empirical (never measured), methodological (no method under constraint Y), contextual (unknown in setting B), or replication (never independently confirmed).

The most common false gap

"Nobody has combined A with B." Reviewers ask why A should help B; if you cannot say, it is not a gap, it is a search result.

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From problem to question to hypothesis

Before	After	What changed
How can we make LLM code generation better?	Does retrieving repository-local type signatures reduce the compile-error rate of LLM-generated Rust functions, compared with no retrieval, at an equal token budget?	Names the intervention, the outcome measure, the comparison and the constraint.
Study the security of IoT devices.	What fraction of the 50 best-selling consumer IoT devices accept firmware updates without signature verification, and does the fraction differ by price tier?	A topic becomes a characterization question with a defined sample.

The hypothesis: falsifiable, directional, with a stop rule

"Retrieval reduces the compile-error rate by at least 20% relative to no retrieval. If the month-2 pilot on 100 functions shows less than 5%, we switch to the fallback: a characterization of which error types retrieval fixes."

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Act are your own reviewer

What a reviewers might say	The fix
"The contribution is unclear."	One-sentence claim in the abstract, repeated at the end of Section 2.
"Related work is a list of papers."	Comparison table; prose organised by assumption, not by paper.
"Why does this matter?"	Heilmeier Q4 as its own paragraph: who, what decision, what changes.
"The evaluation is vague."	Baselines and datasets named; metrics derived from the goal; threats listed.
"This is not feasible in the time."	Milestone plan with a go/no-go gate, compute estimate and buffer.
"No risks are considered."	Risk table with mitigations and fallback contributions.

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